QUALITY ASSURANCE TESTING DOCUMENTATION

(QATD)

Team 166_EyeScream (Domain 1)

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Document Control

Field	Detail
System Under Test (SUT)	UMHackathon 2026 - Team TaxMate
Team Repo URL	https://github.com/lishan0311/TaxMate
Project Board URL	https://trello.com/b/hAykktBk/umhackathon2026-taxmate

Objective:

The primary objective of this QATD is to ensure that TaxMate can reliably handle the end-to-end tax compliance workflow — covering document upload and OCR processing, AI-powered field extraction and anomaly detection, intelligent auto-approval routing, accountant review and digital sign-off, SST form generation, and email notification delivery — with consistent accuracy, graceful failure handling, and acceptable performance under realistic load conditions, with CI/CD checkpoints enforced at integration and deployment stages.

PRELIMINARY ROUND (Test Strategy & Planning)

1. Scope & Requirements Traceability

This section aligns testing back to specific user requirements to ensure every test case is traceable to a defined feature, preventing unplanned feature creep and ensuring no core requirement is left unvalidated.

1.1 In-Scope Core Features

- **Document Upload & OCR Processing:** Tax period selection, batch drag-and-drop upload (PDF, JPG, PNG), PaddleOCR text extraction, Document Repository management, document lock upon owner submission

- **AI Field Extraction & Confidence Scoring:** GLM-powered structured field extraction from unstructured receipt text, per-field confidence scoring, SST ledger classification
- **Anomaly Detection:** Dual-layer rule-based and GLM validation, TIN format checking, SST number validation, amount consistency verification, plain-language anomaly explanations
- **Intelligent Auto-Approval:** Automatic approval for documents at or above 90% confidence threshold; manual review routing for documents below threshold
- **Accountant Review & Sign-off:** Three-panel review workspace, editable AI-extracted fields, Agent Thinking Panel, digital sign-off with handwritten signature pad
- **SST Form Generation:** SST-02 draft PDF with DRAFT watermark, signed SST form upon accountant sign-off, audit trail CSV export
- **Email Notification:** Automated email to accountant upon owner submission; automated email to owner with signed form upon accountant sign-off
- **AI Chatbot & Chat Assistant:** Owner AI Chatbot for tax Q&A; Accountant AI Chat Assistant for in-workspace tax rule queries
- **Statistics Dashboard & AI Insights:** Compliance metrics and GLM-generated insights for both owner and accountant roles

1.2 Out-of-Scope

- Direct API submission to LHDN MyInvois portal
- Full MyInvois E-Invoice generation (55-field JSON output)
- Multi-company or multi-branch account support
- Real-time WhatsApp notification delivery
- Accountant credential verification
- EPF, SOCSO, or payroll module integration

2. Risk Assessment & Mitigation Strategy

Quality assurance risks are anticipatory. The following technical risks are identified based on TaxMate's architecture and requirements, evaluated using the 5×5 Risk Assessment Matrix where Risk Score = Likelihood × Severity.

Technical Risk	**Likelihood (1–5)**	**Severity (1–5)**	**Risk Score (L×S)**	**Mitigation Strategy**	**Testing Approach**
OCR returns empty or incomplete text for low-quality, handwritten, or faded receipts	3	3	9 (Medium)	Flag low-confidence extractions for mandatory human review. Return specific error message to user instead of silent failure. Display receipt quality guidelines at upload screen.	Upload deliberately degraded, handwritten, and low-resolution receipt images. Verify system flags correctly and does not auto-approve or silently fail.
GLM hallucinates a plausible but incorrect TIN or SST registration number	2	5	10 (Medium)	All critical fields pass through rule-based format validators after GLM extraction. Any value failing format validation is flagged before reaching accountant. Dual-layer validation	Inject receipts with deliberately obscured or missing TIN fields. Verify validator catches hallucinated output and routes document to human review queue.

				(rule-based + GLM) prevents single-point failure.	
Document auto-approved at ≥90% confidence despite containing one factually incorrect field	2	5	10 (Medium)	Single flagged field blocks auto-approval for entire document regardless of overall confidence score. Auto-approved documents remain visible in accountant queue for spot-checks. Audit trail records confidence score and approval method for every field.	Engineer a receipt designed to score just above 90% threshold with one subtly incorrect field value. Verify document is NOT auto-approved and is routed to manual review.
Full agent pipeline (OCR → Extract → Anomaly → Generate) fails midway without notifying user	3	4	12 (High)	Three-tier fallback mechanism: retry prompt up to 2 times with reformulated instruction → mark affected fields with low-	Simulate GLM API timeout mid-pipeline. Verify system retries, then flags affected fields, then escalates — without silent

				confidence flag → escalate to human review with detailed error report. No output silently accepted without passing basic rule-based validation.	failure or orphaned document state.
Email notification fails to deliver at critical workflow trigger points (submission / sign-off)	2	3	6 (Medium)	In-platform notification shown alongside email at every trigger point. Notification status visible in both owner and accountant dashboards. Retry mechanism activated for failed email delivery.	Simulate SMTP failure during sign-off trigger. Verify in-platform notification still delivered correctly and email retry mechanism activates.
Accountant review queue becomes unmanageable under high simultaneous	3	3	9 (Medium)	Auto-approval at ≥90% confidence significantly reduces manual review volume. Priority sorting surfaces	Simulate batch submission of 10 documents from multiple client accounts simultaneously. Verify queue

document volume				anomalous and high-value documents first. Accountant dashboard shows cross-client workload visibility.	priority sorting and auto-approval routing behave correctly.
Owner locks submission prematurely before all receipts are uploaded	2	2	4 (Low)	Pre-submission checklist with document count and thumbnail preview shown to owner before lock is confirmed. Accountant can flag incomplete submissions back to owner for re-submission.	Submit with deliberately incomplete document set. Verify pre-lock confirmation prompt appears with correct document count. Verify accountant can flag for owner re-submission.

Risk Assessment Scoring Criteria:

Likelihood (1-5)		*Severity (1-5)*	
1	*Rare*	*1*	*Impact is Negligible*
2	*Unlikely*	*2*	*Impact is Minor*
3	*Possible*	*3*	*Moderate Impact*
4	*Likely*	*4*	*Major Impact*
5	*Almost Certain*	*5*	*Critical Failure of the system*

Risk Score = Likelihood × Severity

Risk Level Reference:

Risk Score	***Risk Level***	***Recommended Action***
1 – 5	*Low*	*Monitor only. Acceptable risks.*
6 – 10	*Medium*	*Mitigate + Testing*
11 – 15	*High*	*Must need mitigating and through testing is required*
16 – 25	*Critical*	*Priority is Highest. Need extensive level of testing.*

3. Test Environment & Execution Strategy

- **Unit Test**
  - Scope: Individual AI agent functions (OCR Agent, Extraction Agent, Anomaly Detection Agent, Generation Agent); rule-based field validators (TIN format, SST registration number format, amount range checks); auto-approval threshold routing logic (≥90% confidence → auto-approve; below threshold → flag for manual review); document lock enforcement after owner submission

  - Execution: Tests written using Pytest with mocked GLM API responses. Execution happens during feature development and is automatically triggered in the CI pipeline upon every push to the GitHub repository.

  - Isolation: External dependencies (Z.AI GLM API, PostgreSQL database, SMTP email server) are fully mocked using pytest-mock. Tests focus solely on the internal logic of each agent function and validator without requiring live API calls or database connections.

  - Pass Condition: Each agent returns the expected structured JSON output for a given input. All invalid TIN and SST number formats are caught and flagged by validators. Documents at or above 90% confidence are correctly routed to auto-approved status. Documents below threshold are correctly flagged for manual review. Edge cases (empty OCR output, missing mandatory fields, zero-amount receipts) return specific error states rather than silent failures.
- **Integration Test**
  - Scope: Full agent pipeline from OCR extraction through to SST form generation (OCR → Extraction → Anomaly Detection → Document Generation); email notification dispatch on owner submission and accountant sign-off events; document lock mechanism preventing post-submission edits from the owner portal; auto-approval routing across the complete pipeline with real receipt inputs

- Execution: Performed after unit tests pass for all involved components. Triggered after merging feature branches — for example, after merging the OCR branch with the Extraction Agent branch. Real API calls made to Z.AI GLM without mocking. Real database writes verified at each pipeline stage.

- Workflow: Real Malaysian receipt files (JPG, PNG, PDF) passed through the full agent pipeline. PostgreSQL database checked after each stage to verify correct document status updates. Email dispatch verified against a configured test SMTP server. Document repository checked to confirm lock state after submission.

- Pass Condition: Full pipeline completes without errors for all tested receipt formats. Extracted fields from real receipts match manually verified ground truth within acceptable margin. Email sent to correct recipient with correct content at each trigger point. Document set is immutable from the owner side after submission lock is confirmed. Auto-approval correctly triggered only when all fields meet the 90% threshold and pass format validation.

- **Test Environment (CI/CD Practice):**
  - Local Testing: Manual testing performed on localhost. React frontend runs on Vite dev server at port 5173. FastAPI backend runs on Uvicorn at port 8000. Vite proxy configured to route /api calls to backend. PostgreSQL 16 database and all dependencies spun up via Docker Compose locally.

  - Staging/CI: GitHub Actions triggers Pytest unit test suite automatically on every push to any branch. Test results visible in GitHub Actions workflow logs. Failed tests block branch from being considered stable.

  - CI/CD Automated Pipeline: GitHub Actions runs basic unit tests at branch level on every push. Full Pytest suite including integration tests is triggered on every Pull Request to the main branch. PR cannot be merged if any test in the suite fails.

- **Regression Testing & Pass/Fail Rules:**

- Execution Phase: Full regression run executed across all in-scope features each time a new branch is merged to main. Covers document upload, OCR extraction, AI field extraction, anomaly detection, auto-approval routing, accountant review, sign-off, SST form generation, and email notification in every regression cycle.

- Pass/Fail Condition: A test case is only considered passed when the actual output perfectly aligns with the expected output. Any mismatch, missing field, incorrect routing, or threshold not achieved results in a Failed status. All failed test cases must be logged in the Defect Log with the corresponding TC ID, description of mismatch, and severity level.

- Continuation Rule: Initial test cases in each phase must meet the passing threshold before testing proceeds to the next phase. Critical test cases — specifically GLM hallucination detection, auto-approval routing, and SST form generation — must achieve 100% pass rate before any deployment gate opens. If these critical cases fail, development team must fix and re-run before continuing.

- **Test Data Strategy:**
  - Manual: 3 mock SME owner accounts created across different business sectors (retail, F&B, services) and 2 mock accountant accounts. Sample Malaysian receipts prepared in three language variants — English only, Bahasa Malaysia only, and mixed English/Chinese — covering normal receipts, receipts with missing fields, receipts with mismatched SST amounts, and deliberately low-quality/handwritten receipts for edge case testing.
  - Automated: SQL seed scripts used to populate the test PostgreSQL database with mock user accounts, pre-created filing periods, and document records at various pipeline stages (uploaded, submitted, under review, signed) to support automated regression runs without manual setup each time.

- **Passing Rate Threshold:**

- A minimum of 85% of all executed test cases must pass overall for the build to be considered stable.
- For critical test cases covering GLM hallucination detection, auto-approval routing logic, document lock enforcement, and SST form generation: 100% must pass with zero exceptions. These functions directly affect the accuracy and legality of compliance output — any failure in production could result in incorrect tax filings or LHDN penalties for SME users.

4. CI/CD Release Thresholds & Automation Gates

This section defines the metrics and thresholds that govern pipeline transitions. If a threshold is not met, the pipeline blocks forward progression.

4.1 Integration Thresholds (Merging to Main)

The main branch must remain a stable source of truth. The following checks are enforced before any merge is accepted.

Checks	**Requirements**	**Project**	**Pass/Failed**
Automatic Build	**Zero Build Error**	**0 Errors (Vite and FastAPI)**	**Passed**
Unit Tests	**100% Passing Rate**	**100%(tests passed for core functions)**	**Passed**
Code Quality	**Zero Linting Error**	**0 Errors (Optimized for functional correctness)**	**Passed**

Test Coverage	**Minimum 81.2%**	**85%**	**Passed**

4.2 Deployment Thresholds (Pushing to Production)

Checks	Requirements	Project	Pass/Failed
Regression Test	**Minimum 90%**	**97%**	**Passed**
AI Output Pass Rate	**Minimum of 80% of documented prompt**	**90%**	**Passed**
Critical Bugs	**Zero P0/01 Bugs**	**0**	**Passed**
API Performance	**Response Time < 800ms**	**30431.8ms**	**Failed**
Security	**API keys are not exposed**	**Clean**	**Passed**

5. Test Case Specifications (Drafts)

Test cases cover the mandatory minimum: 1 Happy Case (full flow), 1 Negative/Edge Case, and 2 NFR (Performance/Load) tests.

Test Case ID	Test Type & Mapped Feature	Test Description	Test Steps	Expected Result	Actual Result
TC-01	Happy Case (Entire Flow): Full Owner-to-Download Workflow	Verify that an SME owner can select a tax period, batch upload receipts, submit to accountant,	1. Owner logs in and selects current tax period. 2. Batch uploads 3 receipts (1 PDF, 1 JPG, 1 PNG). 3. Verifies all 3 appear in Document Repository with	All 3 receipts processed and displayed correctly. Document set locked after	System successfully executed the entire end-to-end workflow: all 3 uploaded formats (PDF, JPG, PNG)

		and successfully download the signed SST form after accountant sign-off — with all modules communicating correctly end-to-end.	correct thumbnails. 4. Clicks Submit — system locks document set. 5. Accountant view pending clients on the dashboard 6. Accountant opens workbench, views queue with 3 documents. 7. High-confidence receipt (≥90%) is shown as auto-approved. 8. Remaining receipts reviewed individually in three-panel workspace. 9. Accountant approves all and provides handwritten digital signature. 10. Signed SST form generated. 11. Owner receives email with signed form. 12. Owner opens Downloads section and downloads signed SST form.	submission — no further edits possible from owner side. Accountant receives email notification within 30 seconds of submission. High-confidence receipt auto-approved correctly. Signed SST form generated and emailed to owner within 30 seconds of accountant sign-off. Owner can download both signed and draft versions	were correctly parsed by the AI Tax Agent with visible thumbnails, and document locking was triggered immediately upon submission to maintain state integrity. The automation engine accurately handled auto-approval for high-confidence receipts (Score: 0.95), while both email notifications and signed SST form generation were completed well within the

				from Downloads section.	30s threshold, averaging 12s per delivery. Finally, data integrity was verified as the signed SST form was successfully retrieved and downloaded from the Owner's portal, confirming seamless cross-module communicatio n. Status: Passed
TC-02	Negative/Edge Case: GLM Hallucinated TIN Caught by Validator	Verify that when GLM extracts a plausible but incorrectly formatted TIN from an obscured receipt, the rule-based validator	1. Prepare receipt with TIN field deliberately obscured or ambiguous. 2. Upload receipt via Owner Portal. 3. Allow full agent pipeline to run. 4. Observe extracted TIN value in document detail	Document is not auto-approved even if overall confidence is above 90%. Anomaly flag "Invalid TIN format" displayed in	Test failed as the system prioritizes the overall confidence score over the rule-based validator; the document was incorrectly auto-approved

		catches the hallucination and routes the document to human review — blocking auto-approval regardless of overall confidence score.	view. 5. Verify TIN format validator result in Agent Thinking Panel. 6. Verify document is routed to manual review queue — not auto-approved. 7. Verify anomaly flag "Invalid TIN format" is displayed with plain-language explanation.	accountant review queue. Plain-language explanation and suggested resolution shown. Document surfaced at top of accountant review queue as high-priority. No silent failure — document status clearly reflects flagged state.	because the GLM output a confidence score of 0.92, which bypassed the TIN format validation check. While the "Invalid TIN format" flag was correctly generated in the backend metadata, it failed to block the auto-approval status or route the document to the manual review queue as expected. This indicates a logic flaw in the decision-making hierarchy where the high-

					confidence threshold from the LLM overrides the hard-coded compliance validators, leading to a silent failure of the safety gate. Status:Failed
TC-03	NFR (Performance): Document Upload API Response Time Under Concurrent Load	Verify that the document upload API endpoint remains responsive within acceptable time limits when multiple users upload simultaneously.	1. Open Postman or load testing tool. 2. Configure 10 concurrent POST requests to /api/documents/upload each with a single JPG receipt file. 3. Execute simultaneous requests. 4. Record average response time and error rate.	Average response time < 800ms across all 10 concurrent requests. 0% error rate — all uploads accepted and queued for processing without failures.	The API endpoint successfully handled the concurrent load with an average response time of 458.6ms across all 10 simultaneous POST requests, well within the 800ms requirement. No request timeouts or 500-series errors were

					observed, maintaining a 0% error rate throughout the burst test. System monitoring via the backend terminal confirmed that the FastAPI asynchronous handlers effectively managed the concurrent file I/O operations, with the longest individual request peaking at 612.4ms during the initial handshake, proving the infrastructure's capability to remain responsive

					under multi-user scenarios. Status: Passed
TC-04	NFR (Load): Maximum Batch Upload Without Cross-Contamination	Verify that uploading the maximum allowed batch of 10 files simultaneously completes without queue failures, data loss, or cross-contamination of extracted fields between documents.	1. Prepare 10 distinct receipts (mix of PDF, JPG, PNG) with known field values for each. 2. Upload all 10 in a single batch drag-and-drop session. 3. Verify all 10 appear in Document Repository. 4. Open each document detail view and verify extracted fields match the correct source receipt. 5. Verify no field values from one receipt appear in another receipt's extraction result.	All 10 documents appear in Document Repository after batch upload. Each document's extracted fields correctly correspond only to its own source receipt. No cross-contamination of field values between documents. Asynchronous processing completes for all 10 without queue failure or timeout.	The batch upload test for 10 concurrent files was successful, achieving a **100% completion rate (Passed)**. During the execution, the system effectively managed simultaneous requests; while 2 out of 10 AI extraction tasks initially encountered temporary GLM API timeouts, the backend's **Retry Logic** automatically intervened and successfully

					re-processed them on the second attempt. No data cross-contamination was found, and all 10 receipts were correctly mapped to their respective database entries. The total processing time for the entire batch remained stable, proving the system's robustness in handling its maximum defined load without manual intervention. Status:Passed

6. AI Output & Boundary Testing (Drafts)

6.1. Prompt/Response Test Pairs

Test ID	Prompt Input	Expected Output (Acceptance Criteria)	Actual Output	Status
AI-01	[e.g. Summarize the 3-paragraph of customer complaint]	[e.g. The output should be only 2-4 sentence, does capture main issue, no fabrication]	Your Test Output	Pass/Fail
AI-02	[e.g. Generate report from a structured Json input]	[e.g. All the JSON fields does reflect accurately and there is no hallucinated values]	Your Test Output	Pass/Fail

Test ID	**Prompt Input**	**Expected Output (Acceptance Criteria)**	**Actual Output**	**Status**
AI-01	Malaysian B2B Tax Invoice (trilingual: English/BM/Chinese) uploaded as JPG via Owner Portal. Invoice from Woodcraft Timber Sdn. Bhd. to Ideal Furniture Sdn. Bhd., Invoice No. INV-SVC-2025-0002, dated 30/03/2025, Total Amount RM 33,000.00, SST	GLM correctly extracts all mandatory fields: Supplier (Woodcraft Timber Sdn. Bhd.), Invoice No (INV-SVC-2025-0002), Date (30/03/2025), SST Registration No (W10-2508-32000001), Total Amount (RM 33,000.00), SST Amount (RM 1,980.00), SST Rate	All mandatory fields successfully extracted. Supplier: Woodcraft Timber Sdn. Bhd. Invoice No: INV-SVC-2025-0002. Date: 30/03/2025. SST Registration No: W10-2508-32000001. Total Amount: RM 33,000.00. SST Amount: RM 1,980.00. SST Rate: 6%. Buyer: Ideal Furniture Sdn. Bhd. SST	Pass

	Amount RM 1,980.00 at 6%.	(6%), Buyer (Ideal Furniture Sdn. Bhd.). SST cross-verification passes — RM 33,000 × 6% = RM 1,980.00. Confidence score ≥ 0.90. Zero risk flags. Document routed to auto-approved. No hallucinated values in any field.	cross-verification passed — calculated SST (RM 33,000 × 6% = RM 1,980.00) matches stated SST amount. Confidence Score: 98%. Risk Flags: None. Approval Status: Auto-Approved by AI.	
AI-02	Same Woodcraft Timber Sdn. Bhd. invoice (INV-SVC-2025-0002) but with a deliberately incorrect SST amount. Invoice states Total Amount RM 33,000.00 but SST Amount stated as RM 3,000.00 — whereas correct SST at 6% should be RM 1,980.00. A deliberate mismatch of RM 1,020.00 between stated and calculated SST.	GLM detects SST amount mismatch. Anomaly flag raised with plain-language explanation: "SST amount mismatch — stated RM 3,000.00, calculated RM 1,980.00 (RM 33,000 × 6%)." Suggested resolution provided. Document NOT auto-approved regardless of other field confidence scores. Confidence score drops below 0.90 due to anomaly. Document surfaced at top of accountant review queue as high-priority.	System successfully detected SST amount inconsistency. Compliance alert raised: "SST amount mismatch — stated amount does not match calculated SST based on taxable supply." GLM reasoning trace in Agent Thinking Panel confirmed tool call sequence: lookup_sst_rule → validate_sst_number → calculate_tax → flag_for_human_review. calculate_tax tool identified discrepancy between stated SST (RM 3,000.00) and expected	Pass

		No hallucinated values in other fields.	SST (RM 1,980.00). Document flagged with anomaly and routed to manual accountant review queue. Auto-approval correctly blocked. Plain-language explanation and resolution suggestion displayed to accountant.	
AI-03	Malaysian F&B receipt uploaded as JPG via Owner Portal. Receipt from Super Kitchen Chilli Pan Mee @ Sri Petaling (Refined Prestige Sdn. Bhd.), SST ID: W10180831036246, Invoice No: 85506, Date: 27/02/2026, Subtotal: RM 97.25, SST 6% inclusive: RM 5.50, Taxable Amount: RM 91.75. Receipt contains trilingual content (English, BM, Chinese characters) across item descriptions.	GLM correctly extracts all mandatory fields despite trilingual mixed-language content: Supplier (Refined Prestige Sdn. Bhd.), SST Registration No (W10180831036246), Invoice No (85506), Date (27/02/2026), Total Amount (RM 97.25), SST Amount (RM 5.50), SST Rate (6% inclusive), Taxable Amount (RM 91.75). Mixed-language item descriptions (Chinese/BM/English) do not cause extraction failure or missing fields. Confidence score	All mandatory fields successfully extracted from trilingual receipt. Supplier: Refined Prestige Sdn. Bhd. (Super Kitchen Chilli Pan Mee @ Sri Petaling). SST Registration No: W10180831036246. Invoice No: 85506. Date: 27/02/2026. Total Amount: RM 97.25. SST Amount: RM 5.50. SST Rate: 6% (inclusive). Taxable Amount: RM 91.75. Mixed-language item descriptions (English/BM/Chinese) extracted without field mapping errors or missing values. Confidence Score: 85%. Risk Flags: 3	Partially Pass

		≥ 0.80. No fields missing due to language switching.	detected — document routed to manual accountant review queue. Auto-approval correctly blocked as confidence score below 90% threshold.	
AI-04	Owner AI Chatbot prompt submitted via Owner Dashboard: "Can I claim SST for my staff lunch expenses?"	Chatbot returns accurate answer in plain language. Correctly states that SST cannot be claimed on staff lunch expenses under Malaysian SST framework. References relevant SST rules. Response factually accurate, concise, and contains no fabricated regulation numbers or hallucinated policy references.	Chatbot correctly responded that SST on staff lunch expenses cannot be claimed. Response explained: (1) SST does not have an input tax credit mechanism unlike GST — Service Tax paid on expenses is a business cost and cannot be recovered; (2) Service Tax on F&B at 8% becomes part of business cost with no offset mechanism against output tax; (3) Even under former GST, staff meal expenses were blocked input tax under Regulation 36 of GST Regulations. Response also surfaced contextual information from the	Pass

			current document session, referencing the previously uploaded receipt and its confidence score and risk flags — demonstrating the chatbot's ability to reference live document context in its answers. Uncertainty note provided regarding income tax deductibility distinction. No hallucinated regulation numbers. Response accurate and well-reasoned.	

6.2. Oversized/Larger Input Test

Fields	Details
Maximum Input Size	[e.g. 1500 Token/800 Word Count/ 3 Page Upload]
Input used while testing	[e.g. 9000 Words - 6x of the limit]
Expected Behavior	[e.g. Reject / Chunking into segments]
Actual Behavior	You test Output
Status	Pass/Fail

Fields	**Details**
Maximum Input Size	10MB per file; maximum 10 files per batch

Input Used While Testing	28MB JPG image file (1.2× the per-file size limit)
Expected Behavior	File rejected before upload completes. Specific error message displayed to user: "File exceeds the 10MB size limit. Please compress or re-upload." No partial processing. No document created in repository.
Actual Behavior	Upload did not complete successfully. System returned a "Request Timeout" error. Document was not created in the repository and no partial processing occurred. However, no specific user-facing error message was displayed to inform the user of the file size limit — the timeout error does not clearly communicate the reason for failure to the user.
Status	**Partial Pass** — Oversized file was correctly rejected and no document was created. However, the error handling is insufficient: a generic timeout message was returned instead of a specific "File too large" message, leaving the user unaware of why the upload failed or how to resolve it.

6.3. Adversarial/Edge Prompt Test

Test Scenario: Upload a completely irrelevant non-receipt image (anime character illustration) as a receipt file via Owner Portal to verify the system handles unrecognisable input gracefully without silent failure or hallucination.

Input: JPG image file containing an illustrated anime character with no transactional data, no supplier information, no monetary amounts, and no SST-related content.

Expected GLM Behaviour:

- PaddleOCR extracts only irrelevant text if any (e.g. "Daddy's Lil Monster" from the t-shirt)
- GLM identifies no recognisable receipt or invoice structure
- System does not attempt to fabricate or hallucinate tax fields from irrelevant content
- Document flagged with error state — not auto-approved
- User sees specific error message such as "Unable to identify a valid receipt or invoice in the uploaded image"
- No supplier name, amount, SST number, or date hallucinated

Actual Behaviour:

- OCR extracted only visible text from image: "Daddy's / PilMonster"
- GLM incorrectly identified "Daddy's / PilMonster" as Supplier Name — this constitutes a hallucination of supplier identity from irrelevant image content
- Total Amount returned as N/A
- Tax Treatment flagged as "UNCLEAR"
- AI Confidence Score: 20.0% — correctly low, indicating GLM recognised the document as unreliable
- System correctly raised 3 Compliance Alerts: No SST registration number found; Document contains no amounts, date, itemized details or TIN; Document does not meet E-Invoice standards
- Document NOT auto-approved due to low confidence score (20% is well below the 90% threshold)
- Agent Thinking Panel correctly executed tool calls: lookup_sst_rule, generate_tax_advice — demonstrating the pipeline continued gracefully without crashing
- No silent failure — document processed to completion with clear anomaly flags

Analysis: The system demonstrated partial adversarial handling. The confidence-based auto-approval gate correctly blocked this document from proceeding. Compliance alerts accurately identified all missing mandatory fields. However, the OCR + GLM pipeline did hallucinate a supplier name ("Daddy's / PilMonster") from irrelevant image text rather than returning a "No valid receipt detected" error state. This indicates the anomaly detection layer successfully caught the invalid document downstream, but an upstream input validation layer that rejects non-receipt images before GLM processing would further strengthen the system.

Status: Partial Pass — Auto-approval correctly blocked; compliance alerts correctly raised; however, supplier name hallucination from irrelevant content identified as an area for improvement.

6.4. Hallucinating Handling

TaxMate mitigates GLM hallucination through a three-layer defence strategy applied at every stage of the agent pipeline.

Layer 1 — Rule-Based Format Validation: After every GLM extraction, all critical fields including TIN, SST registration number, date format, and amount format are passed through rule-based format validators before any downstream processing occurs. Any field value that fails format validation is immediately flagged as an anomaly regardless of the confidence score GLM assigned to it. For example, a hallucinated TIN such as "SG-12345" (incorrect jurisdiction prefix) or "1234" (incorrect length) will fail the TIN format validator and be flagged before the document reaches the accountant.

Layer 2 — Auto-Approval Blocking on Any Anomaly Flag: The 90% confidence auto-approval threshold is applied at document level only when all individual fields pass format validation with no anomaly flags. A single flagged field blocks auto-approval for the entire document regardless of how confidently GLM extracted every other field. This ensures that a single confidently hallucinated value cannot slip through the auto-approval gate undetected.

Layer 3 — Accountant Visibility and Override: All documents including auto-approved ones remain visible in the accountant's review queue with their confidence scores and approval method recorded in the audit trail. Accountants can open and override any auto-approved document at any time. The Agent Thinking Panel streams GLM's full reasoning trace, allowing accountants to identify implausible reasoning steps even when the extracted output appears structurally correct.